

Materials Selection for a Ferritic / Austenitic Stainless Transition Weld with a proposal for a Non-Destructive Testing Method to ensure Volumetric Weld Integrity

**EUR ING Thomas Poon MEng CEng MIMMM MInstNDT
Section Lead – NDE Research & Future Technology
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Introduction to Welding

- The welding of dissimilar austenitic stainless steels to each other is fairly common practice.
- This can be further extended to ferritic and low-alloy steels to create transition or dissimilar metal welds.
- Suitable welding procedures need to be considered with the selection of appropriate filler metals while considering the chemistry of the parent material(s).
- Chemical composition needs careful consideration in the weld region in relation to the parent material(s) to ensure compatibility of mechanical properties.



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Welding Chemistry Considerations

- As a general rule, the weld metal should be at least equal in strength and corrosion properties to the poorest component in the joint.
- The microstructure must be resistant to cracking and a measure of a property known as the Ferrite Number (FN) needs to be considered in relation to this. Hint: Use the Schaeffler diagram.
- The chemistry of the weld filler and associated parent material(s) can be determined by using FN.
- Hint: the chemical composition of the weld is likely to mean that it will be austenitic.



Materials Selection Criteria

- Parent material(s) selection:
 - Select a ferritic carbon alloy with:
 - Yield $> 550\text{MPa}$
 - UTS = $650 - 950\text{MPa}$
 - Elongation $> 15\%$
 - Select an austenitic stainless steel with:
 - Yield $> 200\text{MPa}$
 - UTS $> 550\text{MPa}$
 - Elongation $> 25\%$
- Select filler metal that is appropriate for the welding process you select with a nominal UTS of $> 500\text{MPa}$.



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Material Selection - Solution

- Low carbon alloy steel – SA 508 or similar
- Austenitic stainless steel – Type 305 or similar
- Weld filler – Type 309 or similar

Material	C	Si	Mn	Ni	Cr	Mo	Nb	Cr Equiv	Ni Equiv
SA 508	0.19	0.1	0.25	3.4	1.6	0.45	0.01	2	9
Type 305	0.05	0.25	1.0	10	19.0	0.05	0.05	19	12
Type 309	0.01	0.65	1.0	12	25.0	0.25	0.01	26	13
Material	Yield	UTS	Elong						
SA 508	650 MPa	820 MPa	22%						
Type 305	260 MPa	585 MPa	40%						
Type 309	400 MPa	590 MPa	40%						



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Materials Selection – Solution (Continued)

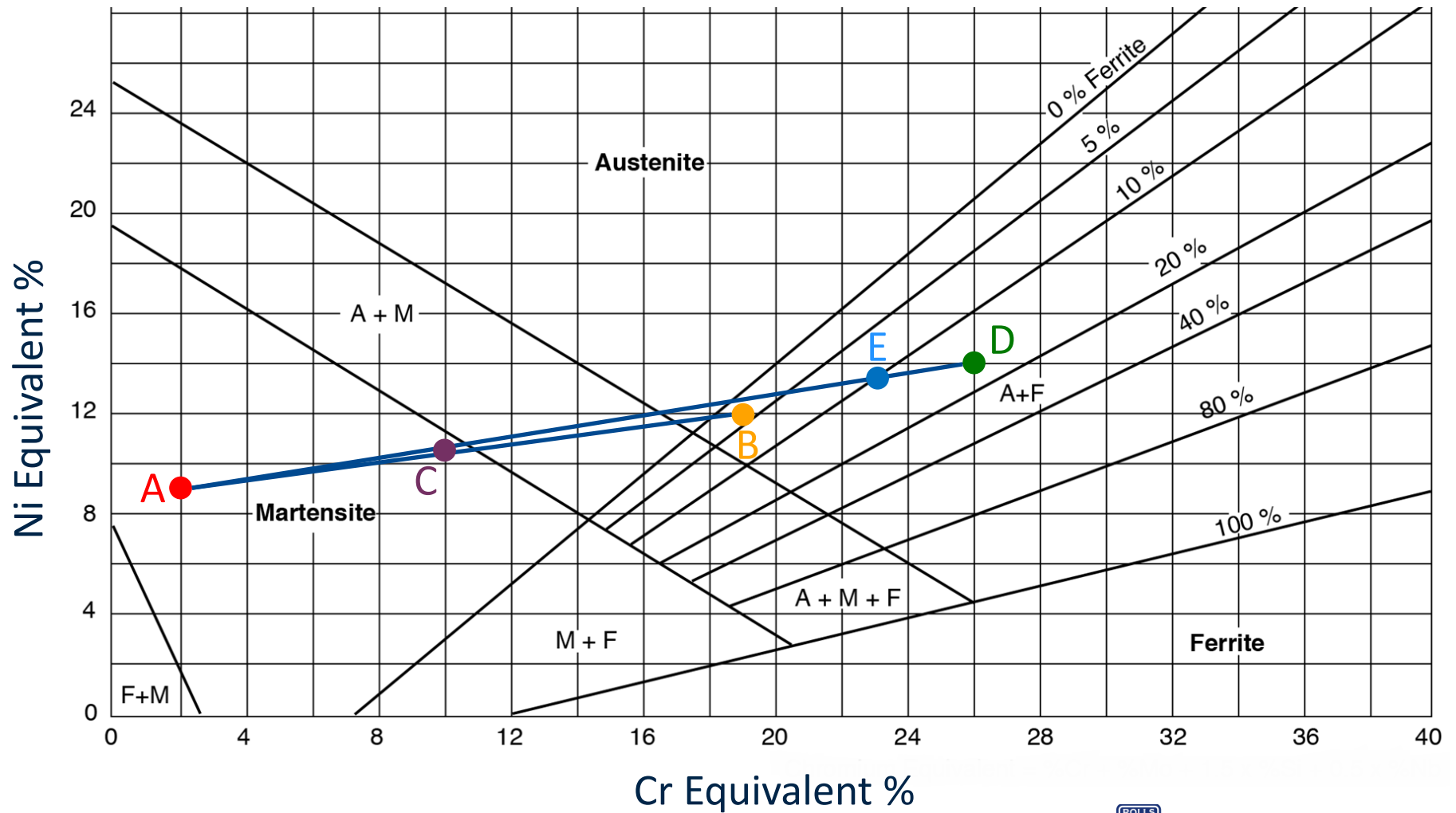
- By using the calculated Cr and Ni equivalence values for each material, these can be plotted on the Schaeffler Diagram (next slide).
- Point A is SA 508, Point B is Type 305. If no weld filler was used then the welded area will have a composition of Point C.
- Point D is the Type 309.
- Point E is the target composition of the weld which is controlled by dilution factor of the two parent metals and the weld filler.



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Shaeffler Diagram

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Materials Selection – Solution (Continued)

- Point C has a FN of 0 and will therefore have very weak crack resistance due to martensite formation.
- This is why an appropriate filler is used, Point D.
- Dilution of the weld filler with the parent material can be controlled to create Point E. In this example the weld with composition Point E would be ideal with FN = 10-11 (acceptable range = 5 – 20). This gives a good balance of crack resistance and corrosion properties.



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Introduction to NDE

- Non-destructive evaluation (NDE) are known by various alternative terms such as; Non-destructive testing (NDT), Non-destructive inspection (NDI), and Non-destructive examination.
- These processes include testing, inspection, and examination, that mainly involves determining a characteristic of an object or to determine whether the object contains irregularities, discontinuities, or flaws.
- The terms irregularity, discontinuity, and flaw can be used interchangeably to mean something that is questionable or of interest in a component.
- A flaw that has been evaluated as rejectable is usually termed a defect.



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Selection of NDE Methods

- Non-destructive evaluation can be classified divided into seven main areas:
 - **Flaw detection and evaluation***
 - Leak detection and evaluation
 - Metrology (measurement of dimension) and evaluation
 - **Location determination and evaluation***
 - Structure or microstructure characterization
 - Estimation of mechanical and physical properties
 - Stress (strain) and dynamic response determination
- ***Primary focus of this case study.**



NDE Introduction (Continued)

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- NDE processes are unlike any other engineering process in that the process itself does not impart any physical added value to the component being tested.
- Unlike a component that is heat treated where the result could be improved mechanical properties or a machining operation that could add value by creating a finished part, NDE does not alter the physical characteristics of the component under test.
- NDE is primarily used to support the safety case for components in critical applications by ensuring that the tested component is fit for service and free from critical defects.



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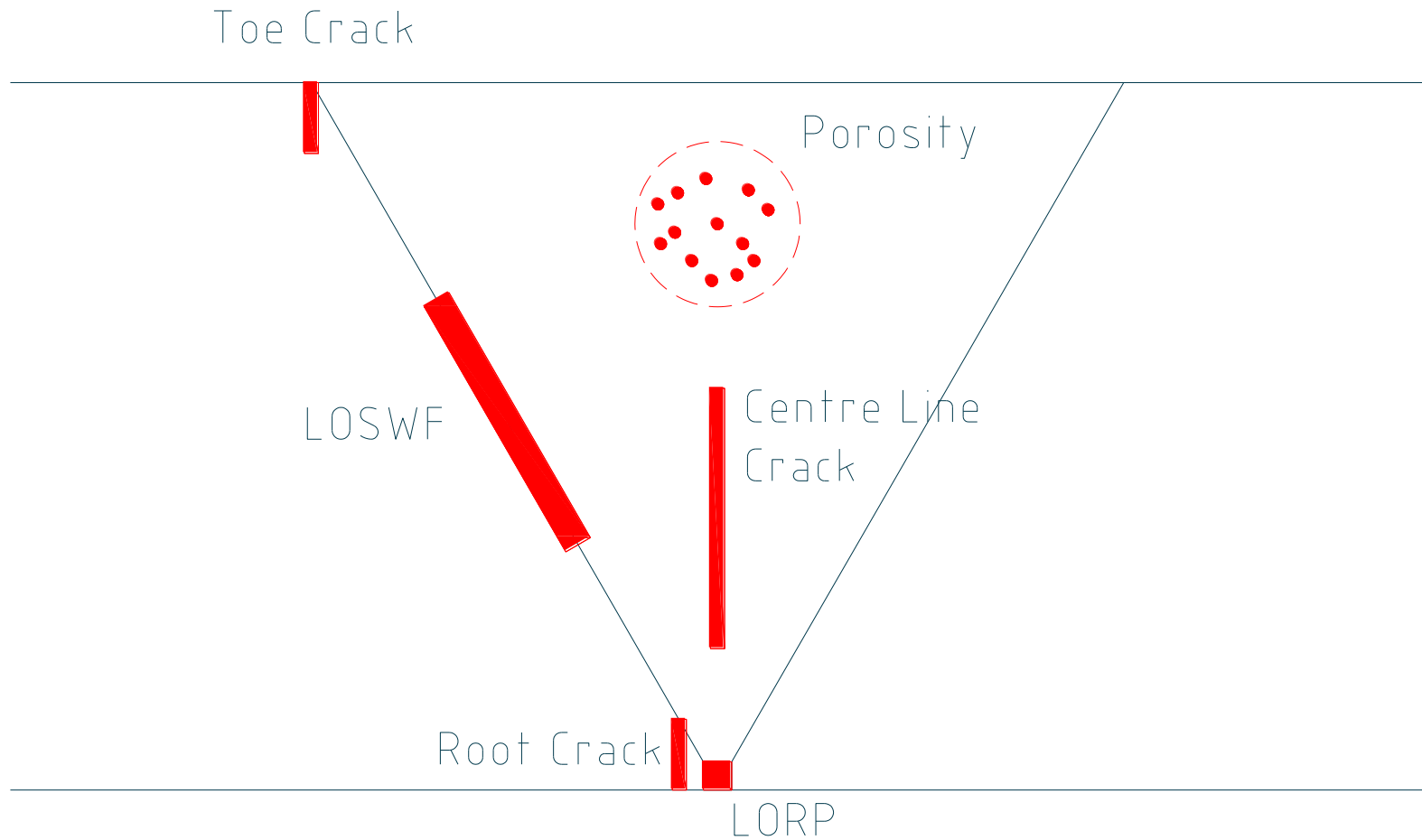
NDE Method Selection Criteria

- The welding defects of interest in this case study are as follows;
 - Lack of Root Penetration (LORP)
 - Lack of Side Wall Fusion (LOSWF)
 - Root Crack
 - Toe Crack
 - Centre Line Crack
 - Porosity



Weld Defects

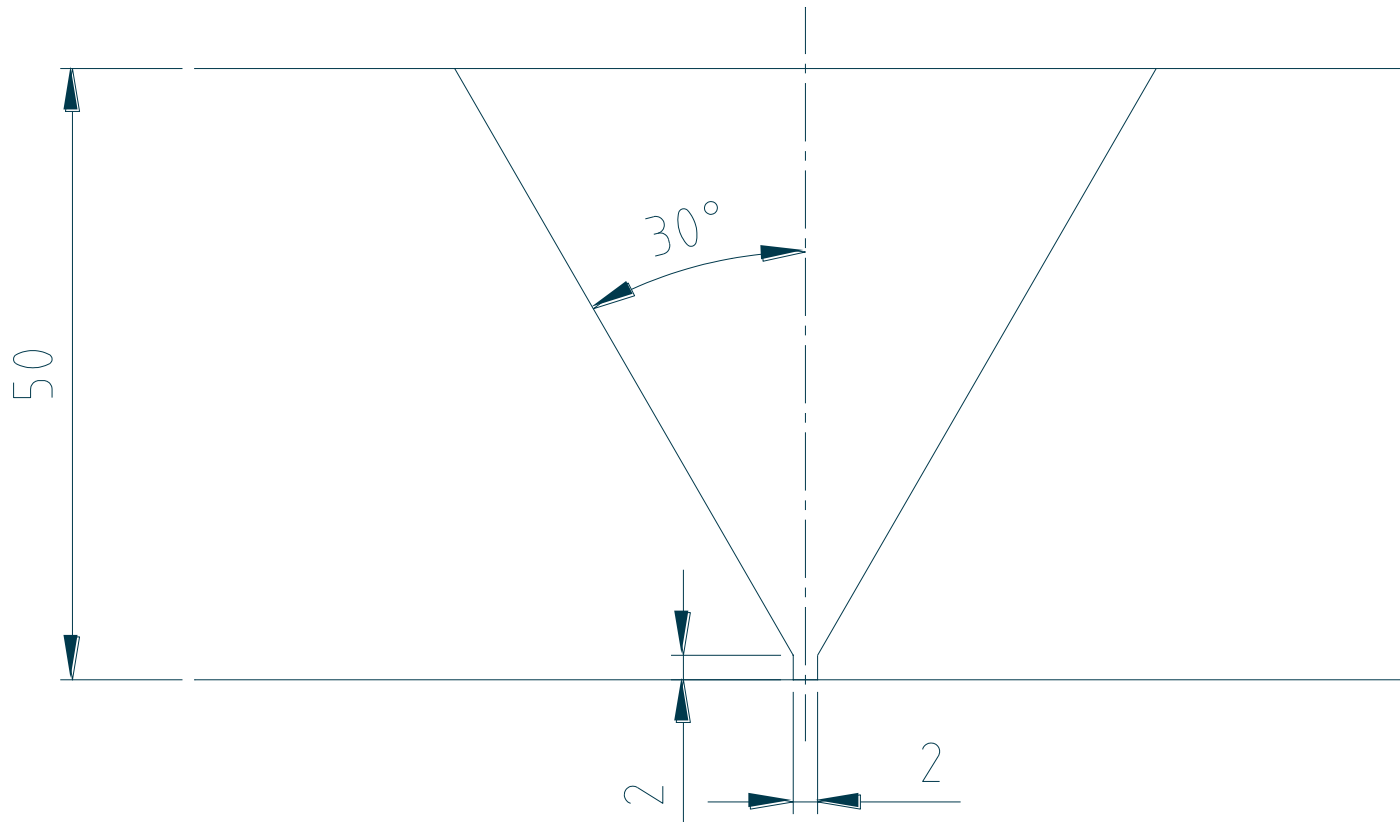
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Case Study Weld Dimensions

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NDE Method / Technique Selection

- Propose an appropriate NDE method to detect the defects by considering the following:
 - The NDE method needs to be a volumetric inspection method that is able to locate and size the defects.
Hint: There are only two, so a 50/50 chance to pick the most appropriate method!
 - The NDE method should be sensitive enough to detect surface defects.
 - Propose specific techniques to detect each of the defects. Hint: CAD modelling approach will help a lot.



NDE Method / Technique Selection

Ultrasonic Testing

- There are 2 volumetric NDE methods. Radiographic Testing (RT) and Ultrasonic Testing (UT).
- UT is considered the more appropriate method of the two because it is superior to RT in detecting surface defects, such as a toe crack.
- UT is a method capable of detecting defects of variable orientations in terms of tilt angle and size.
- UT utilises the propagation of sound waves in the object or material tested. As such, the material properties such as elasticity is critical in determining velocity which must be known in order to locate and size defects appropriately.



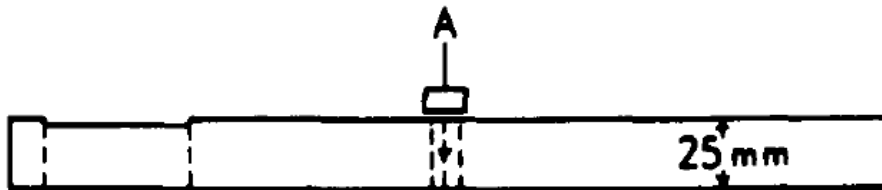
Ultrasonic Testing

Types of Ultrasonic Waves

- There are two wave types normally associated;
 - (a) Longitudinal waves
 - (b) Transverse waves
- The longitudinal waves are produced by a voltage applied to a piezo-electric transducer. In longitudinal waves, the oscillations occur in the longitudinal direction or the direction of wave propagation.
- In the transverse or shear wave, the particles oscillate at a right angle or transverse to the direction of propagation. Shear waves require an acoustically solid material for effective propagation, and therefore, are not effectively propagated in materials such as liquids or gasses.

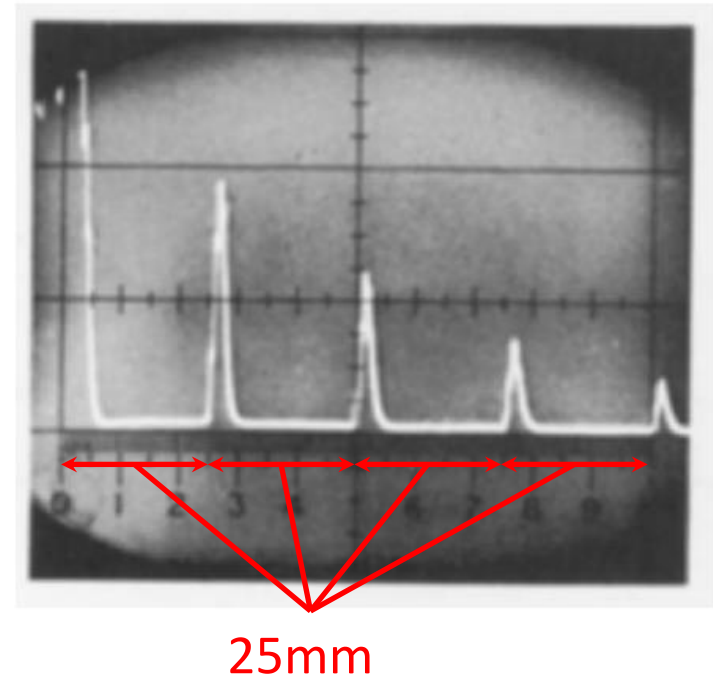


Ultrasonic Testing



Position of probe A gives the 0-100mm thickness range.

- The figure above shows a compression probe firing longitudinal waves through a 25mm thick component.
- The A-scan on the right shows the reflection of sound at the “backwall” or far side.



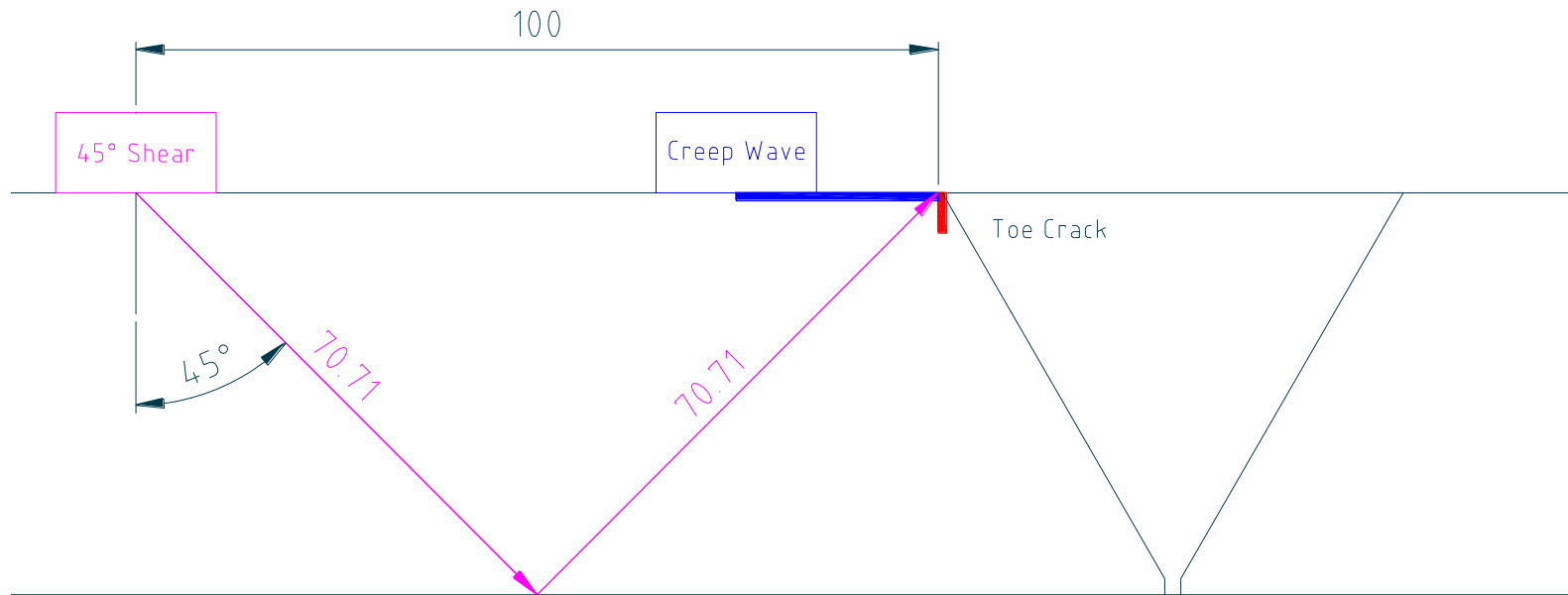
Ultrasonic Testing

- The signals are produced in the previous A-scan figure because they represent an interface reflection from the far side of the component.
- The time base (Y-axis) is calibrated with regards to the material velocity. So, each repeat signal occurs at 25mm increments because the thickness of the component is 25mm.
- Each subsequent signal decreases in amplitude because they represent an additional “return” trip of the sound wave between the near and far surfaces.
- This is the most basic setup of how we use ultrasonic waves, in this case to measure thickness.



Toe Crack Detection

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Two techniques could be used to detect a toe crack.

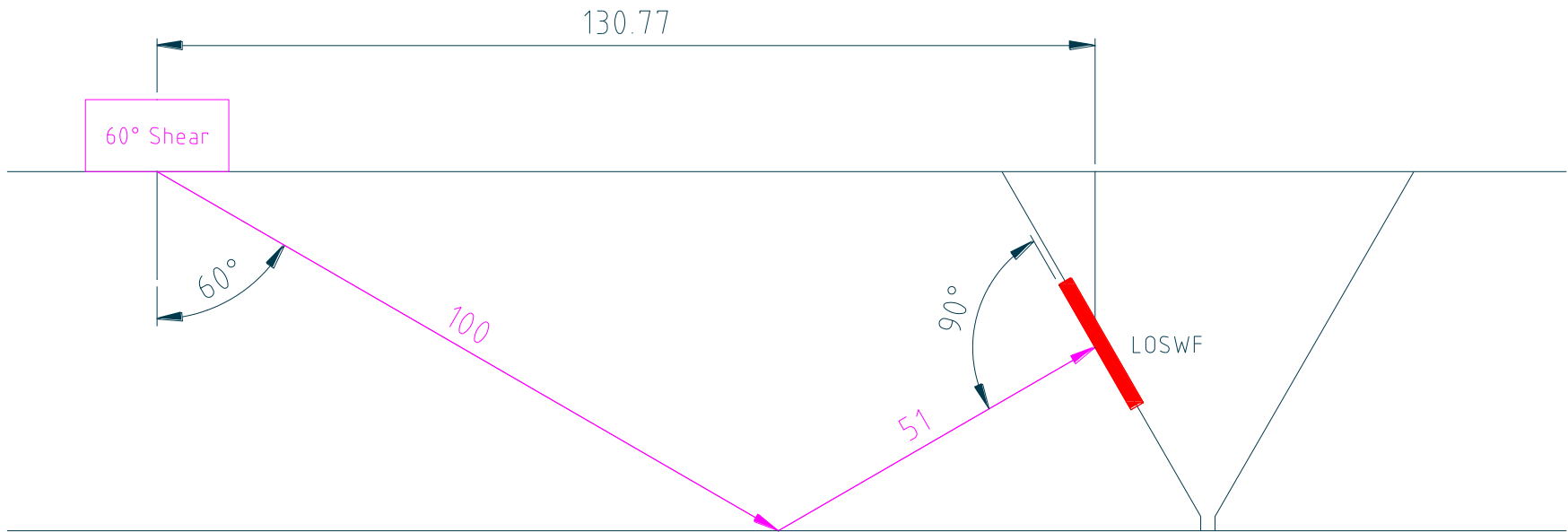
- 1) Creep Wave Probe - creep waves are used, which is a very high angle longitudinal wave (approx 78°) which is an effective surface wave.
- 2) 45° Shear Probe - transverse waves are used from a 45° probe at "full skip".



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LOSWF Detection

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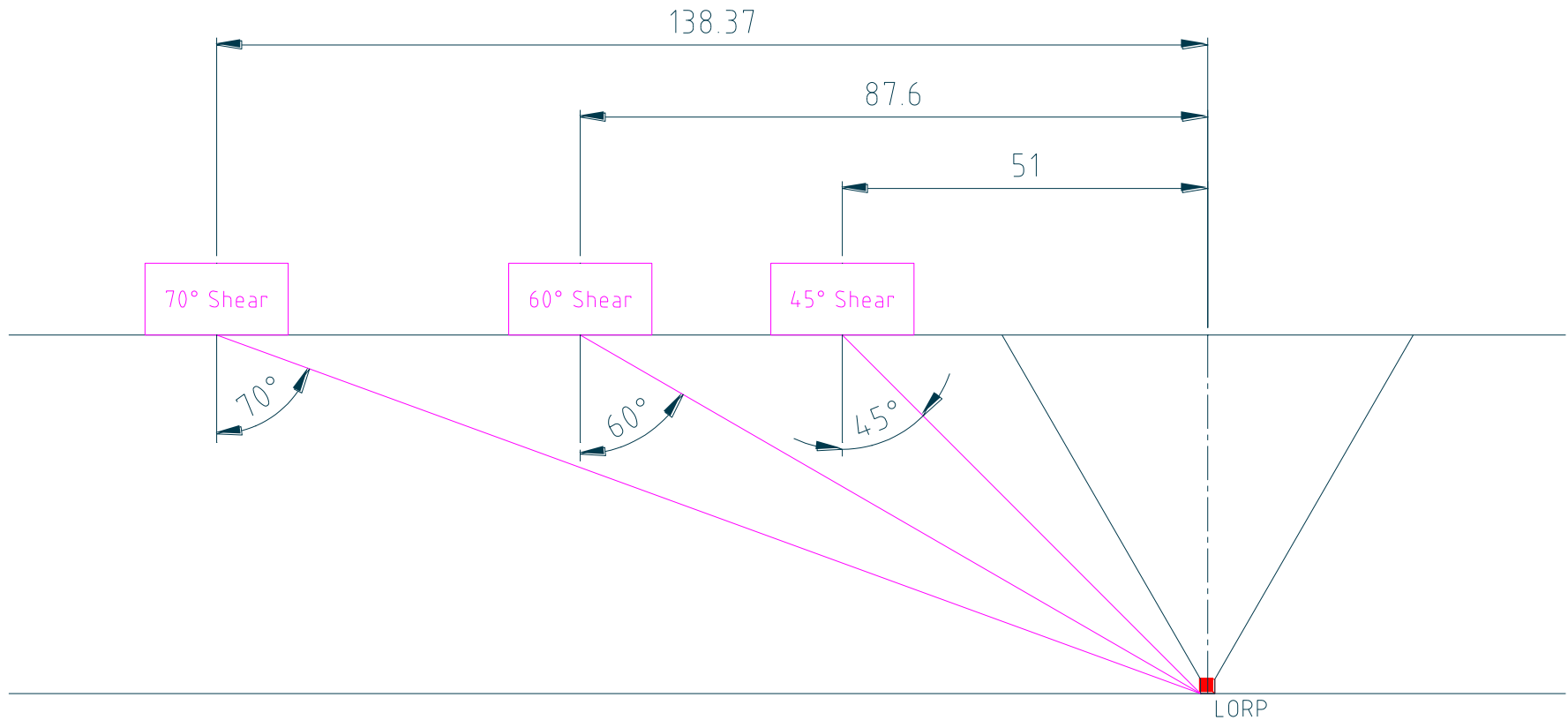
Multiple angled shear wave probes will get a reflective response from a LOSWF defect. However, the angle which will give the best response is a 60° shear probe because the angle of the shear wave will strike the defect at 90° which will give a specular response.



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LORP Detection

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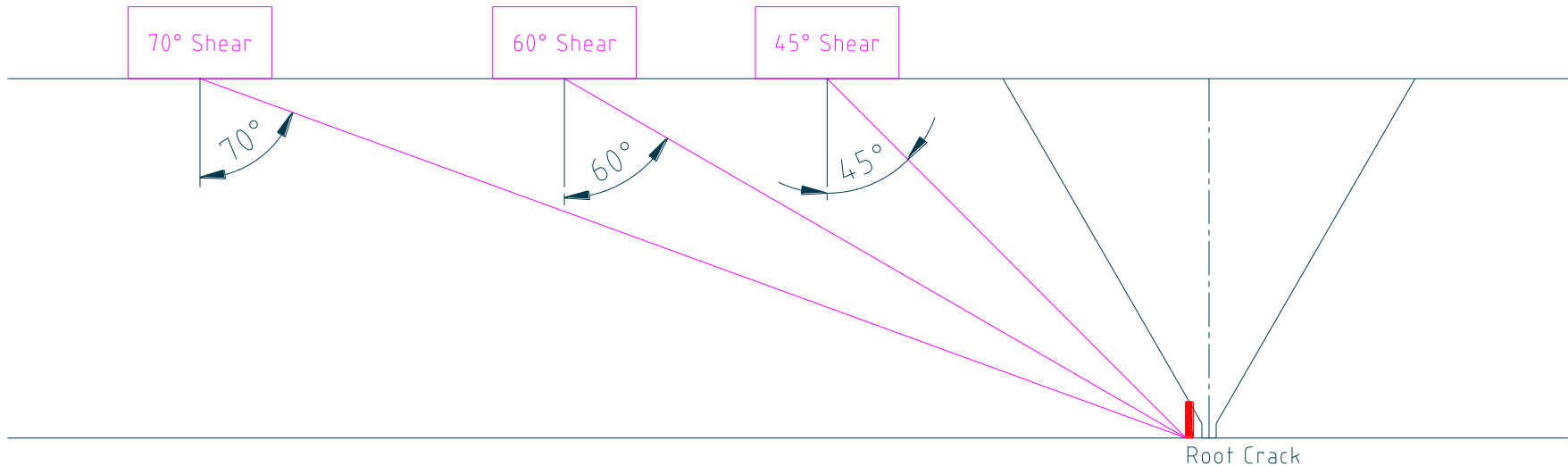
Multiple angled shear wave probes will get a reflective response from a LORP defect as shown above. Due to the nature of the location of such a defect, it should be noted that the defect can be found on either side of the weld centreline.



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Root Crack Detection

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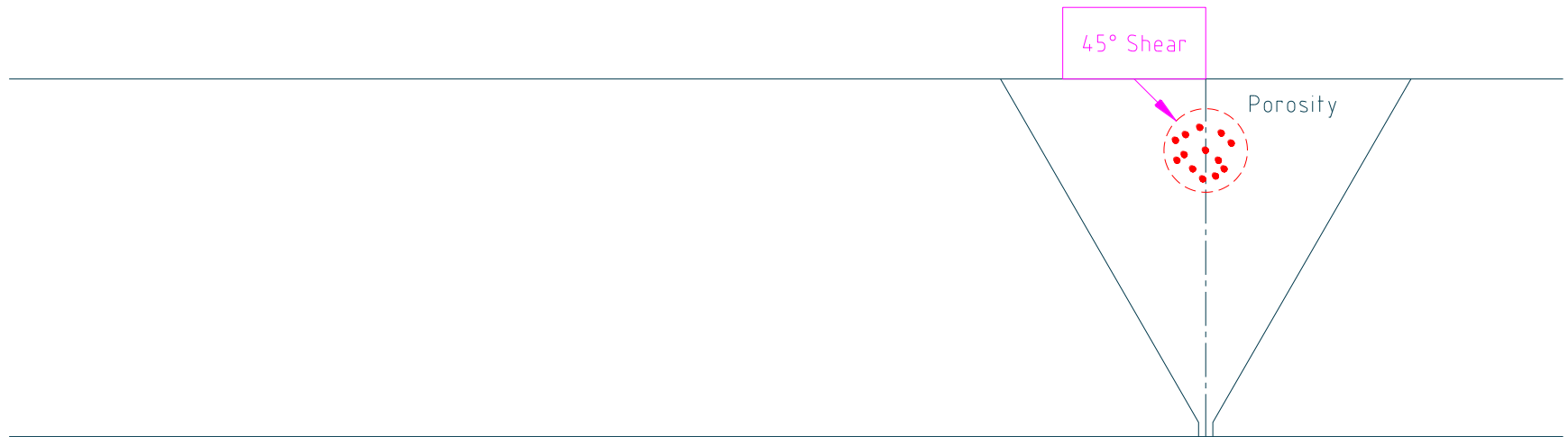
Multiple angled shear wave probes will get a reflective response from a root which is similar to a LORP as shown above. The main difference between a root crack a LORP will be its echodynamic response, which is how the amplitude changes when the probe traverses along the top scanning surface. A LORP will give a large response with a very well defined rise and fall on the timebase when viewed on an A-scan. A crack response will not and seem to have multiple peaks and troughs as would be expected from a relatively rough surface.



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Porosity Detection

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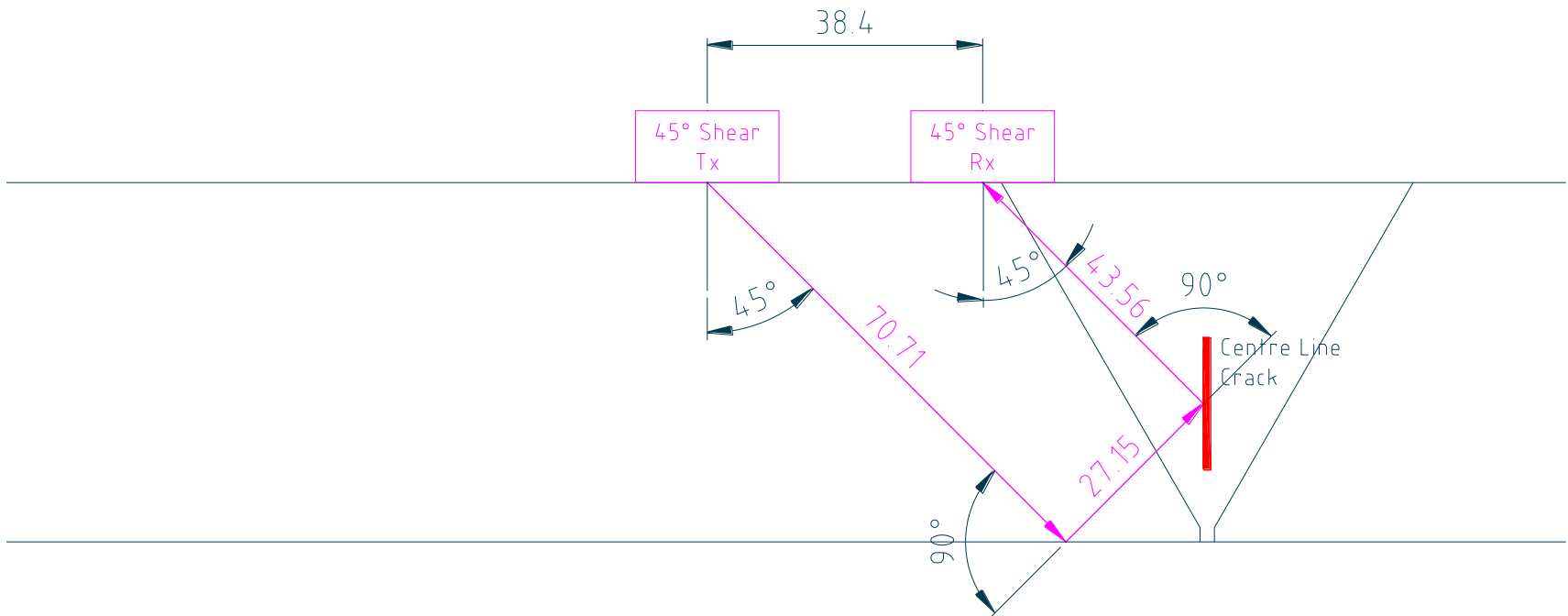
Multiple angled shear wave probes can get a reflective response from porosity cluster. Since the weld cap in this example is machined and accessible by the UT probe, a direct response is possible without skipping off the back surface.



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Centreline Crack Detection

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A centreline crack will not give a specular response from any angled probe. Therefore, a tandem setup with two probes is required. Tx is the transmitting probe and Rx is the receiving probe.



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Case Study Requirements Summary

- Determine chemical composition of parent ferritic, austenitic stainless steel and weld filler metal (50%)
- Select a volumetric NDE method to detect the weld defects described (2%)
- Detail individual techniques to detect each defect (6x8%)



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